Chapter 21 Lipid Biosynthesis

Multiple Choice Questions

1. Which of the following is *not* required in the synthesis of fatty acids?

   A) Acetyl-CoA
   B) Biotin
   C) HCO_3^- (CO_2)
   D) Malonyl-CoA
   E) NADH

Biosynthesis of fatty acids and eicosanoids
Pages: 833–839 Difficulty: 2 Ans: E

2. Which of the following is *not* true of the reaction producing malonyl-CoA during fatty acid synthesis?

   A) It is stimulated by citrate.
   B) It requires acyl carrier protein (ACP).
   C) It requires CO_2 (or bicarbonate).
   D) One mole of ATP is converted to ADP + P_i for each malonyl-CoA synthesized.
   E) The cofactor is biotin.

Biosynthesis of fatty acids and eicosanoids
Pages: 833–834 Difficulty: 2 Ans: B

3. If malonyl-CoA is synthesized from $^{14}CO_2$ and unlabeled acetyl-CoA, and the labeled malonate is then used for fatty acid synthesis, the final product (fatty acid) will have radioactive carbon in:

   A) every C.
   B) every even-numbered C-atom.
   C) every odd-numbered C-atom.
   D) no part of the molecule.
   E) only the omega-carbon atom (farthest carbon from C-1).

Biosynthesis of fatty acids and eicosanoids
Pages: 836–838 Difficulty: 3 Ans: D

4. Which one of the following statements best applies to synthesis of fatty acids in *E. coli* extracts?

   A) Acyl intermediates are thioesters of a low molecular weight protein called acyl carrier protein.
   B) CO_2 or HCO_3^- is essential.
   C) Reducing equivalents are provided by NADPH.
   D) The ultimate source of all the carbon atoms in the fatty acid product is acetyl-CoA.
   E) All of the above

Biosynthesis of fatty acids and eicosanoids

Pages: 833–839 Difficulty: 2 Ans: E

5. In comparing fatty acid biosynthesis with β oxidation of fatty acids, which of the following statements is *incorrect*?

   A) A thioester derivative of crotonic acid (*trans*-2-butenoic acid) is an intermediate in the synthetic path, but not in the degradative path.
   B) A thioester derivative of D-β-hydroxybutyrate is an intermediate in the synthetic path, not in the degradative path.
   C) Fatty acid biosynthesis uses NADPH exclusively, whereas β oxidation uses NAD^+ exclusively.
   D) Fatty acid degradation is catalyzed by cytosolic enzymes; fatty acid synthesis by mitochondrial enzymes.
   E) The condensation of two moles of acetyl-CoA in the presence of a crude extract is more rapid in bicarbonate buffer than in phosphate buffer at the same pH; the cleavage of acetoacetyl-CoA proceeds equally well in either buffer.

Biosynthesis of fatty acids and eicosanoids
Pages: 837–839 Difficulty: 2 Ans: D

6. Which of the following is *not* true of the fatty acid synthase and the fatty acid β-oxidation systems?

   A) A derivative of the vitamin pantothenic acid is involved.
   B) Acyl-CoA derivatives are intermediates.
   C) Double bonds are oxidized or reduced by pyridine nucleotide coenzymes.
   D) The processes occur in different cellular compartments.
   E) The processes occur in the mitochondrial matrix.

Biosynthesis of fatty acids and eicosanoids
Page: 839 Difficulty: 2 Ans: E

7. The rate-limiting step in fatty acid synthesis is:

   A) condensation of acetyl-CoA and malonyl-CoA.
   B) formation of acetyl-CoA from acetate.
   C) formation of malonyl-CoA from malonate and coenzyme A.
   D) the reaction catalyzed by acetyl-CoA carboxylase.
   E) the reduction of the acetoacetyl group to a β-hydroxybutyryl group.

Biosynthesis of fatty acids and eicosanoids
Pages: 840–841 Difficulty: 2 Ans: D

8. Which of the following is *not* true of the fatty acid elongation system of vertebrate cells?

   A) It involves the same four-step sequence seen in the fatty acid synthase complex.
   B) It is located in the smooth endoplasmic reticulum.
   C) It produces stearoyl-CoA by the extension of palmitoyl-CoA.
   D) It uses malonyl-CoA as a substrate.
   E) The immediate precursor of the added carbons is acetyl-CoA.

Biosynthesis of fatty acids and eicosanoids

Pages: 837–839 Difficulty: 2 Ans: E

9. Which of the following is *not* true about precursors required for fatty acid synthesis in animal cells?

   A) NADPH is produced in the cytosol by the pentose phosphate pathway.
   B) NADPH is produced in the nucleus by malic enzyme.
   C) Acetyl-CoA is transported out of the mitochondrion via the citrate shuttle.
   D) CoA is not transported across the mitochondrial membrane.
   E) Malonyl-CoA is formed in the cytosol.

Biosynthesis of fatty acids and eicosanoids
Pages: 839–841 Difficulty: 2 Ans: B

10. Which of these can be synthesized by plants but *not* by humans?

   A) Linoleate [18:2($\Delta^{9,12}$)]
   B) Palmitate (16:0)
   C) Phosphatidylcholine
   D) Pyruvate
   E) Stearate (18:0)

Biosynthesis of fatty acids and eicosanoids
Page: 842 Difficulty: 2 Ans: A

11. The enzyme system for adding double bonds to saturated fatty acids requires all of the following *except*:

   A) a mixed-function oxidase.
   B) ATP.
   C) cytochrome b_5.
   D) molecular oxygen (O_2).
   E) NADPH.

Biosynthesis of fatty acids and eicosanoids
Pages: 842–843 Difficulty: 3 Ans: B

12. Which of these statements about eicosanoid synthesis is *true*?

   A) An early step in the path to thromboxanes is blocked by ibuprofen.
   B) Arachidonate is derived mainly by hydrolysis of triacylglycerols.
   C) Aspirin acts by blocking the synthesis of arachidonate.
   D) Plants can synthesize leukotrienes, but humans cannot.
   E) Thromboxanes are produced from arachidonate via the "linear" path.

Biosynthesis of fatty acids and eicosanoids
Page: 845–847 Difficulty: 3 Ans: A

13. The biosynthesis of triacylglycerols from acetate occurs mainly in:

   A) animals but not in plants.

B) humans after ingestion of excess carbohydrate.
C) humans with low carbohydrate intake.
D) plants but not in animals.
E) None of the above

Biosynthesis of triacylglycerols
Pages: 848–850 Difficulty: 2 Ans: B

14. The synthesis of both glycerophospholipids and triacylglycerols involves:

A) CDP-choline.
B) CDP-diacylglycerol.
C) phosphatidate phosphatase.
D) phosphatidic acid.
E) phosphoethanolamine.

Biosynthesis of triacylglycerols
Page: 849 Difficulty: 2 Ans: D

15. Which of these statements about triacylglycerol synthesis is *correct*?

A) Humans can store more energy in glycogen than in triacylglycerols.
B) Insulin stimulates conversion of dietary carbohydrate into triacylglycerols.
C) It is not a hormone-sensitive process.
D) Mammals are unable to convert carbohydrates into triacylglycerols.
E) Phosphatidate is not on the pathway of triacylglycerol synthesis.

Biosynthesis of triacylglycerols
Pages: 848–849 Difficulty: 2 Ans: B

16. A strategy that is *not* employed in the synthesis of phospholipids is:

A) condensation of CDP-alcohol with diacylglycerol.
B) condensation of CDP-diacylglycerol with alcohol.
C) condensation of CDP-diacylglycerol with CDP-alcohol.
D) exchange of free alcohol with head group alcohol of phospholipid.
E) remodeling of head group alcohols by chemical modification

Biosynthesis of triacylglycerols
Pages: 848–849 Difficulty: 2 Ans: B

17. All glycerol-containing phospholipids are synthesized from:

A) cardiolipin
B) ceramide.
C) gangliosides.
D) mevalonate.
E) phosphatidic acid.

Biosynthesis of membrane phospholipids

Pages: 852–853 Difficulty: 2 Ans: E

18. In *E. coli* the synthesis of phosphatidylethanolamine directly involves:

A) acyl carrier protein.
B) biotin.
C) CDP-choline.
D) phosphatidylglycerol.
E) serine.

Biosynthesis of membrane phospholipids
Pages: 853–854 Difficulty: 2 Ans: E

19. In the synthesis of phosphatidylcholine from phosphatidylethanolamine, the methyl group donor is:

A) a tetrahydrofolate derivative.
B) choline.
C) methanol.
D) *S*-adenosylmethionine (adoMet).
E) serine.

Biosynthesis of membrane phospholipids
Page: 856 Difficulty: 2 Ans: D

20. Palmitoyl-CoA, , is a direct precursor of:

A) cholesterol.
B) malonyl-CoA.
C) mevalonate
D) sphingosine.
E) squalene.

Biosynthesis of membrane phospholipids
Pages: 857–859 Difficulty: 1 Ans: D

21. CDP-diglyceride is *not* involved in the biosynthesis of:

A) phosphatidylcholine.
B) phosphatidylethanolamine
C) phosphatidylglycerol.
D) phosphatidylserine.
E) sphingomyelin.

Biosynthesis of membrane phospholipids
Pages: 853–854 Difficulty: 3 Ans: E

22. Which of the following is *true* of sphingolipid synthesis?

A) All of the carbon atoms of palmitate and serine are incorporated into sphingosine.
B) CDP-sphingosine is the activated intermediate.

C) CO_2 is produced during the synthesis of ceramide from palmitate and serine.
D) Glucose 6-phosphate is the direct precursor of the glucose in cerebrosides.
E) Phosphatidic acid is a key intermediate in the pathway.

Biosynthesis of membrane phospholipids
Page: 857 Difficulty: 2 Ans: C

23. Which of the following is *not* an intermediate in the synthesis of lanosterol from acetyl-CoA?

A) Isopentenyl pyrophosphate
B) Malonyl-CoA
C) Mevalonate
D) Squalene
E) β-Hydroxy-β-methylglutaryl-CoA (HMG-CoA)

Biosynthesis of cholesterol, steroids and isoprenoids
Pages: 859–863 Difficulty: 2 Ans: B

24. Cholesterol is synthesized from:

A) acetyl-CoA.
B) choline.
C) lipoic acid.
D) malate.
E) oxalate.

Biosynthesis of cholesterol, steroids and isoprenoids
Page: 859 Difficulty: 1 Ans: A

25. A 30-carbon precursor of the steroid nucleus is:

A) farnesyl pyrophosphate.
B) geranyl pyrophosphate.
C) isopentenyl pyrophosphate.
D) lysolecithin.
E) squalene.

Biosynthesis of cholesterol, steroids and isoprenoids
Pages: 859–863 Difficulty: 1 Ans: E

26. Which of these statements about cholesterol synthesis is *true*?

A) Cholesterol is the only known natural product whose biosynthesis involves isoprene units.
B) Only half of the carbon atoms of cholesterol are derived from acetate.
C) Squalene synthesis from farnesyl pyrophosphate results in the release of two moles of PP_i for each mole of squalene formed.
D) The activated intermediates in the pathway are CDP-derivatives.
E) The condensation of two five-carbon units to yield geranyl pyrophosphate occurs in a "head-to-head" fashion.

Biosynthesis of cholesterol, steroids and isoprenoids

Pages: 859–863 Difficulty: 2 Ans: C

27. Which of the following is derived from a sterol?

A) Bile salts
B) Gangliosides
C) Geraniol
D) Phosphatidylglycerol
E) Prostaglandins

Biosynthesis of cholesterol, steroids and isoprenoids
Page: 864 Difficulty: 2 Ans: A

28. Chylomicrons carry _____ in the _____.

A) triacylglycerols; cell
B) triacylglycerols; blood
C) cholesterols; blood
D) fatty acids; blood
E) fatty acids; cell

Biosynthesis of cholesterol, steroids and isoprenoids
Page: 865 Difficulty: 2 Ans: B

29. Lipoprotein particles in human blood do *not* contain:

A) an apolipoprotein B isoform.
B) cholesterol.
C) cholesteryl esters.
D) lecithin.
E) triglycerides.

Biosynthesis of cholesterol, steroids and isoprenoids
Page: 865 Difficulty: 2 Ans: D

30. Which of the following contains the highest percentage of cholesteryl esters?

A) Chylomicrons
B) VLDL
C) LDL
D) HDL
E) None of the above contain *any* cholesteryl esters.

Biosynthesis of cholesterol, steroids and isoprenoids
Page: 865 Difficulty: 1 Ans: C

31. Which of the following is *not* a step in the uptake of cholesterol from the bloodstream into cells?

A) LDL containing ApoB-100 is recognized by the LDL receptor.
B) LDL bound to the LDL receptor leads to endocytosis of the complex.
C) The cholesteryl esters in the LDL are hydrolyzed to release free cholesterol.

D) The internalized receptor is degraded.
E) The ApoB-100 protein is degraded.

Biosynthesis of cholesterol, steroids and isoprenoids
Page: 868 Difficulty: 2 Ans: D

32. Which of these statements about the regulation of cholesterol synthesis is *not* true?

A) Cholesterol acquired in the diet has essentially no effect on the synthesis of cholesterol in the liver.
B) Failure to regulate cholesterol synthesis predisposes humans to atherosclerosis.
C) High intracellular cholesterol stimulates formation of cholesteryl esters.
D) Insulin stimulates HMG-CoA reductase.
E) Some metabolite or derivative of cholesterol inhibits HMG-CoA reductase.

Biosynthesis of cholesterol, steroids and isoprenoids
Pages: 869–871 Difficulty: 2 Ans: A

33. Which of the following does *not* contribute to lowering the risk of atherosclerosis?

A) Statin drugs
B) HDL
C) Reverse cholesterol transport
D) ApoA-I
E) Foam cells

Biosynthesis of cholesterol, steroids and isoprenoids
Pages: 871–874 Difficulty: 2 Ans: E

34. Which of these compounds is *not* synthesized by a pathway that includes isoprene precursors?

A) Natural rubber
B) Plastoquinone
C) Vitamin A
D) Vitamin B_{12}
E) Vitamin K

Biosynthesis of cholesterol, steroids and isoprenoids
Pages: 871–874 Difficulty: 2 Ans: E

Short Answer Questions

35. If you start with acetyl-CoA labeled with ^{14}C in the methyl-carbon, where does the label end up in malonyl-CoA?

Ans: Malonyl-CoA would be labeled in C-2 (C-1 is the acyl carbon).

Biosynthesis of fatty acids and eicosanoids
Page: 834 Difficulty: 1

36. If you perform fatty acid biosynthesis with CO_2 labeled with ^{14}C, where does the label end up in a fatty acid?

Ans: There will not be any radioactivity in any fatty acid because the carboxyl group that is added to form malonyl-CoA is lost in the first condensation reaction.

Biosynthesis of fatty acids and eicosanoids
Page: 835 Difficulty: 2

37. The reaction sequence that leads to fatty acid synthesis includes (1) condensation, (2) first reduction reaction, (3) dehydration, and (4) second reduction. Show the first reduction reaction, with any required cofactors.

Ans: β-ketoacyl-ACP + NADPH + $H^+ \longrightarrow$ β–hydroxyacyl-ACP + $NADP^+$

(See Fig. 21-2.)

Biosynthesis of fatty acids and eicosanoids
Page: 835 Difficulty: 2

38. Fatty acid synthesis and fatty acid breakdown occur by similar pathways. Describe, very briefly, four ways in which the synthetic and breakdown pathways differ.

Ans: Fatty acid synthesis (in any order) (1) employs NADPH as reducing agent; (2) involves an acyl group bound to a protein, ACP; (3) takes place in the cytosol of animals; (4) involves the condensation of malonyl- and acetyl-groups; (5) involves the formation of the D-β-hydroxyacyl derivative. Fatty acid breakdown (1) employs NAD^+ as electron acceptor; (2) involves acyl groups bound to coenzyme A; (3) occurs in the mitochondrial matrix; (4) does not involve malonyl-derivatives; (5) involves the L-stereoisomer of the β-hydroxyacyl derivative.

Biosynthesis of fatty acids and eicosanoids
Pages: 834–838 Difficulty: 2

39. The synthesis of fatty acids and their breakdown by β-oxidation occur by separate pathways. Compare the two paths by filling in the blanks below. (Some blanks may require more than one answer.)

	Synthesis	β-oxidation
Activating group	_____	_____
Electron carrier coenzyme(s)	_____	_____
Basic units added or removed	_____	_____
Cellular location of process	_____	_____

Ans:

	Synthesis	β-oxidation
Activating group	acyl carrier protein	CoA—SH
Electron carrier coenzyme(s)	NADPH	NAD^+
Basic units added or removed	malonyl- and acetyl-	acetyl-
Cellular location of process	cytosol in animals,	mitochondrial matrix

chloroplast in plants

Biosynthesis of fatty acids and eicosanoids
Pages: 834–838 Difficulty: 2

40. Show the structure of each intermediate in the conversion of β-hydroxybutyryl-ACP to butyryl-ACP by the fatty acid synthetase complex. Show where cofactors participate. In your first intermediate, circle the carbon atoms that are derived from malonyl-CoA.

    Ans: β-hydroxybutyryl-ACP is first dehydrated, yielding *trans*-Δ^2-butenoyl-ACP, which is then reduced to butyryl-ACP, with NADPH as the reducing agent. (See Fig. 21-6.) The carbon atoms coming from malonyl-CoA are those closer to the ACP.

Biosynthesis of fatty acids and eicosanoids
Page: 837 Difficulty: 3

41. Describe the mechanism for moving acetyl-CoA produced in the mitochondrial matrix into the cytosol for fatty acid synthesis.

    Ans: Acetyl-CoA in the mitochondrial matrix condenses with oxaloacetate to form citrate in a reaction catalyzed by citrate synthase. Citrate moves out of the matrix via the citrate transporter. Citrate in the cytosol is cleaved by citrate lyase, yielding acetyl-CoA and oxaloacetate. To complete the cycle, oxaloacetate in the cytosol is reduced to malate, which moves into the mitochondrial matrix on the malate-α-ketoglutarate transporter, and is converted to oxaloacetate in the matrix.

Biosynthesis of fatty acids and eicosanoids
Pages: 840–841 Difficulty: 3

42. Explain briefly why we require fats in our diets.

    Ans: Dietary fats provide the linoleate and linolenate that we need (for eicosanoid synthesis) but cannot synthesize.

Biosynthesis of fatty acids and eicosanoids
Pages: 842–845 Difficulty: 2

43. Sketch the pathway from arachidonate to thromboxanes and explain how aspirin blocks the synthesis of thromboxanes.

    Ans: (See Fig. 21-15.) Aspirin blocks the activity of cyclooxygenase (COX).

Biosynthesis of fatty acids and eicosanoids
Pages: 846–847 Difficulty: 3

44. Explain the triacylglycerol cycle and how drugs such as rosiglitazone (Avandia) promote this cycle and help lower fatty acid levels in the bloodstream.

    Ans: The triacylgycerol cycle is the constant cycling of fatty acids from adipose tissue to liver and back again. It begins with hydrolysis of fatty acids in adipose tissue followed by transport of the free fatty acids through the bloodstream for re-esterification in the liver. Once in the liver, the process is reversed and the fatty acids end up in adipose tissue again. (See Fig. 21-22.) The constant attaching/detaching

of fatty acids requires a relatively high concentration of glycerol-3-phosphate, which is synthesized from pyruvate in a "truncated" gluconeogenic pathway. (See Fig. 21-21.) Drugs such as rosiglitazone promote higher levels of PEP carboxykinase, thus raising the concentration of glycerol-3-phosphate available for re-esterification of free fatty acids in adipose tissue.

Biosynthesis of triacylglycerols
Pages: 850–852 Difficulty: 3

45. Describe two basic strategies for activating precursors in the biosynthesis of phospholipids.

Ans: (1) Activate the head group by attachment to CDP, as in CDP-choline, then displace CMP with the hydroxyl group of glycerol in diacylglycerol. (2) Activate diacylglycerol by the attachment of CDP, then displace CMP with the hydroxyl group of the head group.

Biosynthesis of membrane phospholipids
Pages: 853–855 Difficulty: 2

46. Show the two different biosynthetic pathways from phosphatidic acid to phosphatidylcholine. You may use shorthand notation for phosphatidic acid, but name any cofactors required in the path and show where they are involved.

Ans: Phosphatidic acid condenses with CTP to form CDP-diacylglycerol; then serine displaces CMP, yielding phosphatidylserine. Decarboxylation of phosphatidylserine yields phosphatidylethanolamine, and methylation of the amino group of phosphatidylethanolamine with three methyl groups, each donated by *S*-adenosylmethionine, yields phosphatidylcholine. (See Figs. 21-25, and Fig. 21-27.) In an alternative path, phosphocholine condenses with CTP, forming CDP-choline; phosphatidic acid is dephosphorylated to yield diacylglycerol; and diacylglycerol displaces CMP from CDP-choline, forming phosphatidylcholine. (See Fig. 21-28.)

Biosynthesis of membrane phospholipids
Pages: 854–856 Difficulty: 2

47. Describe briefly the four stages in the pathway from acetyl-CoA to lanosterol.

Ans: (1) synthesis of mevalonate from acetate; (2) conversion of mevalonate to two activated isoprenes (isopentenyl pyrophosphate and dimethylallyl pyrophosphate); (3) condensation of six activated isoprene units to form squalene; (4) cyclization of squalene to form the steroid nucleus of lanosterol. (See Fig. 21-33.)

Biosynthesis of cholesterol, steroids, and isoprenoids
Pages: 860 Difficulty: 2

48. Show the reaction that limits the rate of cholesterol synthesis from acetate, indicating the role of any cofactors that participate.

Ans: This is the reaction catalyzed by HMG-CoA reductase. NADPH is the cofactor. (See Fig. 21-34.)

Biosynthesis of cholesterol, steroids, and isoprenoids
Pages: 861 Difficulty: 2

49. Show the pathway from acetyl-CoA to mevalonate, indicating the roles of any cofactors.

Ans: (See Fig. 21-34.)

Biosynthesis of cholesterol, steroids, and isoprenoids
Page: 861 Difficulty: 3

50. Show the pathway from mevalonate to dimethylallyl pyrophosphate, indicating where any cofactors participate.

Ans: (See Fig. 21-35.)

Biosynthesis of cholesterol, steroids, and isoprenoids
Page: 861 Difficulty: 3

51. The synthesis of cholesterol begins with the condensation of a four-carbon unit from _____ with the two carbons of acetyl-CoA to form a six-carbon derivative. Give its structure, and circle the atoms that originated in the acetyl-CoA.

Ans: Acetoacetyl-CoA. The six-carbon product is *β*-hydroxy-*β*-methylglutaryl-CoA (HMG-CoA). (The structure and carbons originating in acetyl-CoA are shown in Fig. 21-34.)

Biosynthesis of cholesterol, steroids, and isoprenoids
Page: 861 Difficulty: 3

52. Describe the formation of farnesyl pyrophosphate from activated isoprenyl units.

Ans: This is the reaction in which dimethylallyl pyrophosphate condenses head-to-tail with isopentenyl pyrophosphate, with the elimination of PP_i. (See Fig. 21-36.)

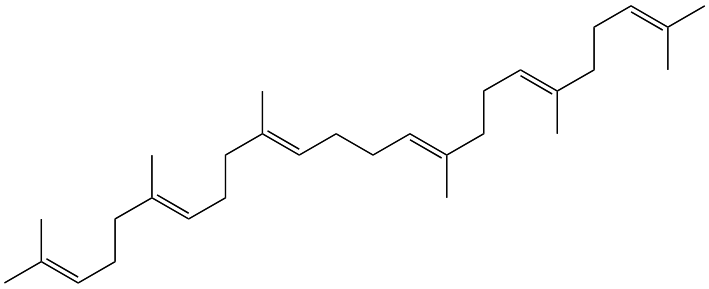
Biosynthesis of cholesterol, steroids, and isoprenoids
Page: 862 Difficulty: 2

53. Show the structure of isopentenyl pyrophosphate and of dimethylallyl pyrophosphate. Connect with a dotted line the two carbon atoms that will be joined when these two molecules condense to form the 10-carbon intermediate in cholesterol biosynthesis.

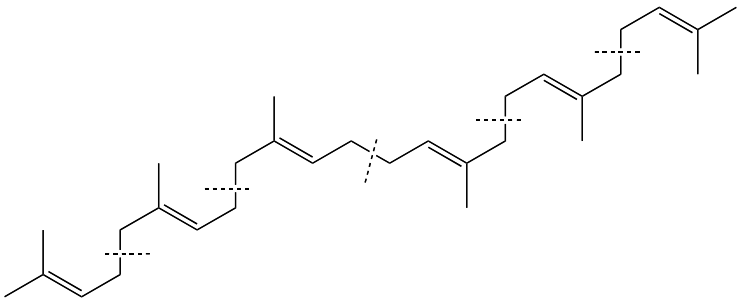
Ans: The "tail" of isopentenyl pyrophosphate (its methylene carbon) displaces pyrophosphate from the "head" of dimethylallyl pyrophosphate, forming geranyl pyrophosphate. (See Fig. 21-36.)

Biosynthesis of cholesterol, steroids, and isoprenoids
Page: 862 Difficulty: 3

54. Shown below is a molecule of squalene, which is composed of 6 isoprene units. Draw lines to indicate the junctions between the six isoprene units.



Ans:



Biosynthesis of cholesterol, steroids, and isoprenoids
Page: 862 Difficulty: 2

55. When a 10-carbon unit and a 5-carbon unit condense to a 15-carbon intermediate in the pathway to cholesterol, the mechanism of condensation is basically different from that of the condensation of two 15-carbon units to the 30-carbon compound squalene. Describe the two condensations in enough chemical detail to illustrate the difference between them. (You should show relevant structures.)

    Ans: The condensation of the 10-carbon geranyl pyrophosphate with the 5-carbon isopentenyl pyrophosphate is a "head-to-tail" condensation; the methylene carbon (tail) of isopentenyl pyrophosphate attacks and is joined to the "head" (the carbon that bears the pyrophosphate group) of geranyl phosphate. The condensation of two molecules of farnesyl pyrophosphate is "head-to-head"; both pyrophosphate groups are eliminated, and the carbons to which they were attached are joined, forming squalene. (See Fig. 21-36.)

Biosynthesis of cholesterol, steroids, and isoprenoids
Page: 862 Difficulty: 3

56. What are plasma lipoproteins? What is their general role in mammalian metabolism?

    Ans: Plasma lipoproteins are protein-lipid aggregates that circulate in the blood, carrying phospholipids, triacylglycerols, cholesterol, and cholesteryl esters from their points of synthesis or absorption to the tissues in which they will be used. The lipid/protein ratio, and therefore the density, of plasma lipoproteins varies, giving rise to particles separable by ultracentrifugation: HDL, LDL, VLDL, and chylomicrons, for example.

Biosynthesis of cholesterol, steroids, and isoprenoids
Pages: 864–866 Difficulty: 2

57. Describe the process by which cholesterol esters in the bloodstream enter cells.

Ans: The process is receptor-mediated endocytosis (described in Fig. 21-41). LDL particles containing cholesterol esters interact via apoB-100 with specific LDL receptors in the cell surface, initiating endocytosis that brings the LDL into the cell within endosomes. Endosomes fuse with lysosomes, and lysosomal enzymes degrade the apoproteins; cholesterol esters are released into the cytosol, and LDL receptors recirculate, reappearing on the cell surface.

Biosynthesis of cholesterol, steroids, and isoprenoids
Page: 868 Difficulty: 2

58. Describe (briefly) two classes of genetic defects in humans that could produce an elevated blood serum cholesterol level.

Ans: (1) Mutations in the LDL receptor result in the failure of cells to take up by receptor-mediated endocytosis the cholestrol in LDL; the resulting high level of LDL in the blood is characteristic of familial hypercholesterolemia. (2) Mutations in the ABC1 protein of HDL result in a failure of the HDL to take up cholesterol and remove it from the blood.

Biosynthesis of cholesterol, steroids, and isoprenoids
Pages: 871-873 Difficulty: 2

59. The synthetic compound mevinolinic acid, also called lovastatin, is a potent competitive inhibitor of HMG-CoA reductase (hydroxymethylglutaryl-CoA reductase). Predict and explain the effect of this drug on serum cholesterol levels in humans.

Ans: HMG-CoA reductase catalyzes the rate-limiting step in cholesterol biosynthesis. By inhibiting this step, lovastatin reduces the endogenous production of cholesterol and lowers the level of cholesterol in the blood.

Biosynthesis of cholesterol, steroids, and isoprenoids
Page: 872 Difficulty: 1